Answer **one** question from this Section in the spaces provided.

8 **(a)** A horizontal string is stretched between two fixed points A and B. A vibrator is used to oscillate the string and produce an observable stationary wave.

At one instant, the moving string is straight, as shown in Fig. 8.1.

Fig. 8.1

The dots in Fig. 8.1 represent the positions of the nodes on the string. Point P on the string is moving downwards.

The wave on the string has a speed of 35 m s^{-1} and a period of 0.040 s.

(i) Explain how the stationary wave is formed on the string.

....

....

....

.... [2]

(ii) On Fig. 8.1, sketch a line to show a possible position of the string a quarter of a cycle later than the position shown in Fig. 8.1. [1]

(iii) Determine the horizontal distance from A to B.

distance = ... m [2]

(iv) A particle on the string has zero displacement at time $t = 0$. From time $t = 0$ to time $t = 0.060$ s, the particle moves through a total distance of 72 mm.

Calculate the amplitude of oscillation of the particle.

amplitude = ... mm [2]

(b) Light from a laser is used to produce an interference pattern on a screen as shown in Fig. 8.2.

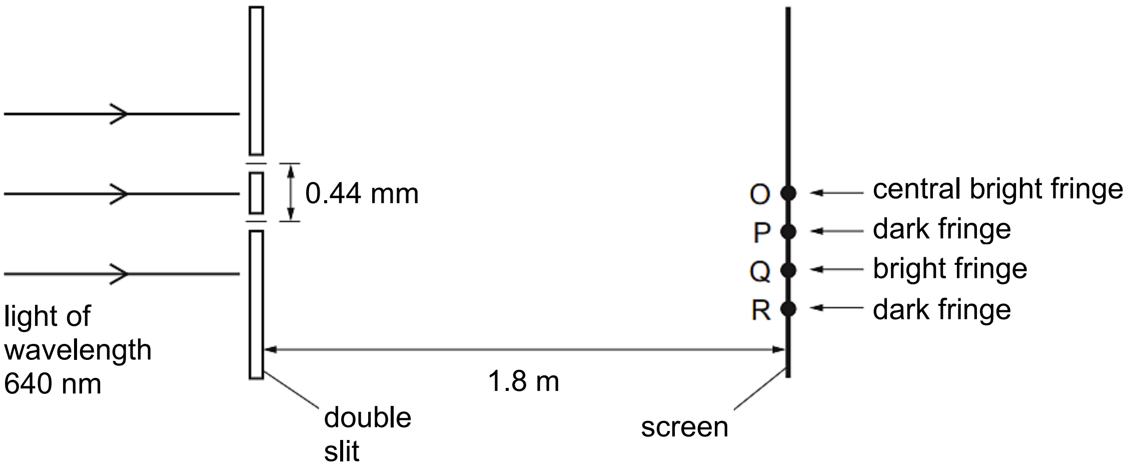

Fig. 8.2 (not to scale)

The light of wavelength 640 nm is incident normally on two slits that have a separation of 0.44 mm. The double slit is parallel to the screen. The perpendicular distance between the double slit and the screen is 1.8 m.

The central bright fringe on the screen is formed at point O. The next dark fringe below point O is formed at point P. The next bright fringe and the next dark fringe below point P are formed at points Q and R respectively.

(i) The light waves from the two slits are coherent.

State what is meant by coherent.

.... [1]

(ii) For the two light waves superposing at R, calculate

1. the difference in their path lengths from the slits,

path difference = ... nm [1]

2. their phase difference.

phase difference = ... ° [1]

(iii) Determine the distance OR.

distance OR = ... m [2]

(iv) The intensity of the light incident on the double slit is increased without changing the frequency.

Describe how the appearance of the fringes after this change is different from, and similar to, their appearance before the change.

....

....

....

....

.... [3]

(v) The light of wavelength 640 nm is now replaced by blue light from a laser.

State and explain the change, if any, that must be made to the separation of the two slits so that the fringe separation on the screen is the same as it was for light of wavelength 640 nm.

....

....

....

.... [2]

(c) A diffraction grating is used with different wavelengths of visible light. The angle θ of the fourth-order maximum from the zero-order (central) maximum is measured for each wavelength.

The variation with wavelength λ of $\sin\theta$ is shown in Fig. 8.3.

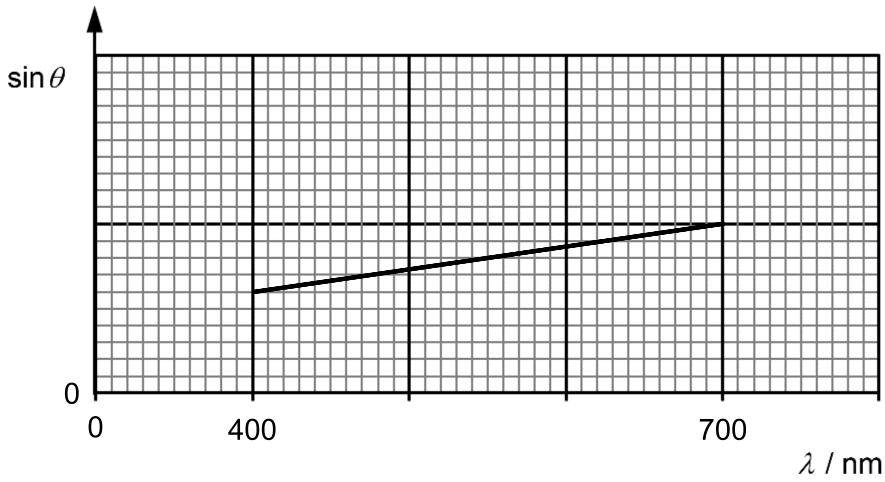

Fig. 8.3

(i) The gradient of the graph is G.

Determine an expression, in terms of G, for the distance d between the centres of two adjacent slits in the diffraction grating.

d = ... [1]

(ii) On Fig. 8.3, sketch a graph to show the results that would be obtained for the second-order maxima. [2]

9 (a) Explain what is meant by the *half-life* of a nuclide.